



SECJ3623 MOBILE APPLICATION PROJECT

PROJECT PROPOSAL

INVENTORY MANAGEMENT SYSTEM

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Project Proposal

Project Title

Inventory Management System (IMS)

A Mobile Application for Engineering Store Management

1. Introduction

This project proposes the development of an Inventory Management System (IMS), a mobile application designed to improve inventory operations at the FNBN engineering store. The system is intended to replace the current manual and semi digital practices with a centralized and reliable digital platform.

At present, inventory records are maintained using stock cards, Excel spreadsheets, and paper documentation. These methods often result in delayed updates, data inconsistency, limited traceability, and a high dependency on manual effort. In a maintenance driven environment where machine availability is critical, such limitations can directly affect operational efficiency.

The IMS aims to digitalize inventory management by providing real time stock visibility, accurate tracking of spare part usage, and seamless linkage between inventory transactions, work orders, and assets.

2. Project Background

The FNBN engineering store plays a vital role in supporting maintenance and operational activities by ensuring the availability of spare parts and consumables. However, the existing inventory management approach relies heavily on manual processes that are prone to human error and inefficiency.

Stock updates are often delayed, physical records may be misplaced, and tracing the usage history of a spare part can be time consuming. These challenges make it difficult to maintain accurate stock levels and support timely procurement decisions.

The proposed Inventory Management System addresses these issues by introducing a mobile based solution that centralizes inventory data and automates key processes such as stock updates, part issuance, and alert generation.

3. Project Objectives

The objectives of the Inventory Management System are as follows:

1. To eliminate manual and paper based record keeping in spare part management.
2. To provide real time visibility of inventory levels, part locations, and usage.
3. To ensure full traceability of spare parts linked to maintenance work orders and assets.
4. To reduce machine downtime through faster and more accurate part issuance.
5. To support proactive procurement through low stock alerts and data driven reorder decisions.

4. Scope of the Project

The scope of the IMS project focuses on the design and development of a mobile application that supports core inventory operations within the FNBN engineering store.

The system will include the following functional scope:

- User authentication with role based access control.
- Management of spare part master data and storage locations.
- Inventory issuance and receiving using barcode scanning.
- Purchase order creation and tracking.
- Real time stock monitoring and reporting.
- Automatic alerts for low stock and critical spare parts.

The project scope covers system analysis, design, and prototype level implementation. Full enterprise integration and large scale deployment are considered outside the scope of this proposal.

5. System Users and Stakeholders

Stakeholders

The key stakeholders involved in this project include:

- Engineering Department
- Maintenance Team

- Store Management
- FNBN Operations Management

System Users

The system is designed to support two primary user roles.

Admin or Store Supervisor

This role is responsible for managing item master data, monitoring stock levels, creating purchase orders, generating reports, and overseeing overall system usage.

Storekeeper or Technician

This role focuses on day to day inventory operations such as issuing and receiving spare parts, scanning barcodes, viewing item details, and linking parts to work orders or assets.

6. Key Features

Admin and Store Supervisor Features

- Secure login with role based dashboard access.
- Management of item master list and storage locations.
- Creation and tracking of purchase orders.
- Monitoring of stock levels with reporting capabilities.
- Receipt of low stock and critical spare alerts.

Storekeeper and Technician Features

- Secure login and mobile friendly interface.
- Barcode scanning for issuing and receiving parts.
- Access to item details, stock balance, and storage location.
- Ability to link issued parts to specific work orders or assets.
- Viewing of transaction history and traceability records.

7. Development Approach and Sprint Planning

The project will be developed using an iterative approach divided into structured sprints to ensure gradual and controlled system development.

Sprint 1 focuses on project setup, user interface design, and backend configuration.

Sprint 2 covers user authentication and role based dashboard implementation.

Sprint 3 includes inventory management functionality, barcode integration, and purchase order management.

Sprint 4 involves reporting features, alert mechanisms, system testing, and deployment preparation.

This approach allows early validation of system functionality and continuous improvement based on feedback.

8. Expected Outcomes and Benefits

The implementation of the Inventory Management System is expected to deliver the following benefits:

- Improved accuracy and consistency of inventory records.
- Faster access to real time stock information.
- Enhanced traceability of spare parts across maintenance activities.
- Reduced machine downtime due to timely part availability.
- Better decision making supported by reliable inventory data.

9. Conclusion

The proposed Inventory Management System offers a practical and effective solution to the challenges currently faced in managing the FNBN engineering store inventory. By replacing manual processes with a centralized mobile based system, IMS enhances visibility, traceability, and operational efficiency.

Through real time stock updates, barcode enabled transactions, and automated alerts, the system supports proactive inventory control and reliable maintenance operations. Overall, the IMS aligns with FNBN's digital transformation goals and provides a strong foundation for more efficient and accountable engineering store management.